

**IN THE UNITED STATES DISTRICT COURT
FOR THE DISTRICT OF DELAWARE**

QORVO, INC.,	)	
	)	
Plaintiff,	)	
	)	Case No. 1:21-cv-01417-JPM
v.	)	
	)	
AKOUSTIS TECHNOLOGIES, INC. and	)	
AKOUSTIS, INC.,	)	
	)	
Defendants.	)	

**ORDER DENYING PLAINTIFF’S MOTION TO EXCLUDE OPINIONS OF
CAROLYN IRWIN AND DEFENDANTS’ MOTION TO EXCLUDE OPINIONS OF
MELISSA BENNIS**

Before the Court are Plaintiff’s Motion to Exclude Expert Testimony of Ms. Carolyn Irwin; Defendants’ Reply; and Plaintiff’s Response. (ECF Nos. 453-54, 479, 492.) Also before the Court are Defendants’ Motion to Exclude Expert Testimony of Melissa Bennis, Plaintiff’s Reply, and Defendants’ Response. (ECF Nos. 456-57, 482, 491.) Parties appeared for oral argument on these issues on April 2, 2024, and filed supplemental letters on the exclusion of Ms. Bennis. (ECF Nos. 522-23, 525.) For the reasons discussed below, Plaintiff’s Motion to Exclude the Testimony of Carolyn Irwin is **DENIED**. Defendants’ Motion to Exclude the Testimony of Melissa Bennis on “Head-Start” Damages is **DENIED**. Defendant’s Motion to Exclude the Testimony of Melissa Bennis on poaching damages will be addressed by separate order, and Parties are **ORDERED** to submit supplemental briefing as detailed below.

I. BACKGROUND

This is a civil action for alleged patent infringement, false advertising, unfair and deceptive trade practices, and other related claims. Qorvo filed a complaint in this action on October 4, 2021. (ECF No. 1.) A First Amended Complaint (“FAC”) and Second Amended Complaint (“SAC”) were filed on February 18, 2022 and February 8, 2023 respectively. (ECF Nos. 28, 125.) Akoustis filed their Answer to the SAC on March 15, 2023. (ECF No. 158.)

Qorvo alleges patent infringement of one or more claims of U.S. Patent No 7,522,018 (the “’018 Patent”) and U.S. Patent No. 9,735,755 (the “’755 Patent”) by Akoustis. (ECF No. 125.) Qorvo also alleges that Akoustis engaged in unfair and deceptive trade practices under N.C. Gen. Stat. § 75-1, misappropriation of trade secret under 18 U.S.C. § 1832 and N.C. Gen. Stat. § 66-152, violations of the Racketeer Influenced and Corrupt Organizations Act (“RICO”) under 18 U.S.C. § 1961, and civil conspiracy under state law. (*Id.*) Qorvo also brings false advertising claims under 15 U.S.C. § 1125(a) and false patent marking claims under 35 U.S.C. § 292 relating to U.S. Patent No. 10,256,786 (the “’786 Patent”). (*Id.*) Qorvo’s RICO and false patent marking claims were dismissed at the summary judgment stage. (ECF No. 545.)

II. LEGAL STANDARD

In general, witnesses may only testify in the form of an opinion if the opinion is “rationally based on the witness’s perception; [h]elpful to clearly understanding the witness’s testimony or to determining a fact in issue; and [i]s not based on scientific, technical or other specialized knowledge within the scope of Federal Rule of Evidence 702.” FED. R. EVID. 701. Federal Rule of Evidence 702 provides that a “witness who is qualified as an expert by knowledge, skill, experience, training, or education may testify in the form of an opinion or otherwise[,]” so long as the proponent of that testimony demonstrates to the court that it is more likely than not that “the expert’s scientific, technical, or other specialized knowledge will help

the trier of fact to understand the evidence or to determine a fact in issue; []the testimony is based on sufficient facts or data; []the testimony is the product of reliable principles and methods; and []the expert’s opinion reflects a reliable application of the principles and methods to the facts of the case.” FED. R. EVID. 702.

The trial court “ensur[es] that an expert’s testimony both rests on a reliable foundation and is relevant to the task at hand.” Bayer Healthcare LLC v. Baltaxa Inc., Case No. 16-cv-1122, 2019 WL 330149, *1 (D. Del. Jan. 25, 2019) (quoting Daubert v. Merrell Dow Pharms., Inc., 509 U.S. 597, 594 (1993)). The Third Circuit has framed Rule 702 as “embod[ying] a trilogy of restrictions on expert testimony: qualification, reliability, and fit.” Schneider v. Fried, 320 F.3d 396, 404 (3d Cir. 2003) (citing In re Paoli Railroad Yard PCB Litigation, 35 F.3d 717, 749 (3d Cir. 1994) (citing Daubert, 509 U.S. at 579)). “[A] broad range of knowledge, skills, and training qualify an expert[,]” so long as the witness possesses specialized expertise. (Id.) For testimony to be reliable, it “must be based on the ‘methods and procedures of science’ rather than on ‘subjective belief or unsupported speculation’; the expert must have ‘good grounds for his or her belief.’” Paoli, 35 F.3d at 742. When determining whether proposed expert testimony is reliable, a district court may consider:

(1) Whether a method consists of a testable hypothesis; (2) whether the method has been subject to peer review; (3) the known or potential rate of error; (4) the existence and maintenance of standards controlling the technique’s operation; (5) whether the method is generally accepted; (6) the relationship of the technique to methods which have been established to be reliable; (7) the qualifications of the expert witness testifying based on the methodology; and (8) the non-judicial uses.

Schneider, 320 F.3d at 405 (citing Paoli, 35 F.3d at 742 n.8 (internal citations omitted)). “Fit” determines whether “the proffered connection between the scientific research or test result to be presented and particular disputed factual issues in the case” is sufficiently strong. Paoli, 35 F.3d at 743. “Fit” requires more than bare logical relevance. Id. at 386. Rather, “fit asks

whether the proffered testimony is sufficiently helpful.” Insight Equity v. Transitions Optical, Inc., 252 F. Supp. 3d 382, 386 (D. Del. 2017).

III. ANALYSIS

a. *Melissa Bennis*

i. *Head-Start Damages*

Defendants argue that Melissa Bennis’ opinions on “head-start” damages are unreliable because “revenue is not a cognizable basis for calculating unjust enrichment except as part of the calculation of a defendant’s ‘ill-gotten’ profits[.]” (ECF No. 457 at PageID 20009.) Defendants also argue that even if revenue were an adequate basis, Ms. Bennis “errs by including in her revenue calculation all of Akoustis’ historical revenue, including revenue unrelated to any alleged use of Qorvo’s purported trade secrets.” (*Id.* at PageID 20010.)

Plaintiffs counter that Ms. Bennis’ “head-start” damages methodology is well-established, and that the use of revenue as a substitute for value is necessitated because Akoustis did not earn a profit during the time period in question. (ECF No. 482.) Plaintiffs also argue that the distinction between the “head-start” methodology Defendants use to describe Ms. Bennis’ work and the time value methodology actually used by Ms. Bennis is significant, and point to use of time value methodologies in non-trade-secret cases. (ECF No. 482 (citing U.S. v. Allegheny Ludlum Corp., 366 F.3d 164, 178 (3d Cir. 2004); U.S. v. Torlai, 728 F.3d 932, 945 n.11 (9th Cir. 2013); Raithborne Land Co. v. Ascent Energy, Inc., 610 F.3d 249, 257 (5th Cir. 2010)).)

One of the forms of damages provided under the Defend Trade Secrets Act are damages for “any unjust enrichment caused by the misappropriation of the trade secret that is not addressed in computing damages for actual loss.” 18 U.S.C. § 1836(b)(3)(B). North Carolina’s

trade secret law provides the same. N.C. Gen. Stat. § 66-154(b). As Akoustis acknowledges, unjust enrichment damages based on a head start theory can be based “either in a defendant’s profits or its incremental increases in business value realized from alleged wrongdoing.” (ECF No. 491.); Sabre GLBL, Inc. v. Shan, 779 F. App’x 843 (3d Cir. 2019).

While Ms. Bennis’ method lacks a testable hypothesis and Plaintiffs have not shown it is subject to peer review, it is well-based in accepted techniques. See Schneider, 320 F.3d at 405 (citing Paoli, 35 F.3d at 742 n.8 (internal citations omitted)). Head-start damages based on profits are accepted as a reliable methodology for DTSA damages. See Sabre GLBL, Inc. v. Shan, 779 F. App’x 843 (3d Cir. 2019). The only distinction between Ms. Bennis’ technique and that accepted by other courts is the substitution of revenue as an input for profit or incremental increase in business value. Ms. Bennis’ qualifications to opine on this technique are undisputed: Ms. Bennis has an M.B.A. in accounting from Northwestern University’s Kellogg School of Management, is a licensed CPA, and is a leader in the Illinois CPA Society. (ECF No. 458.) The question is, then, whether there is a known or potential rate of error for the use of revenue to estimate valuation, whether there are standards controlling this substitution’s operation, whether the method is generally accepted, and any non-judicial uses. See Schneider, 320 F.3d at 405 (citing Paoli, 35 F.3d at 742 n.8 (internal citations omitted)). Here, the Sedona Conference text cited by Defendants as a source of generally accepted knowledge in the field states that “[i]n the case of a start-up . . . the company’s valuation is closely tied to the value of its trade secret technology, and the company’s investment value may be tantamount to or at least closely related to the investment value of the trade secret.” (ECF No. 458.) This source therefore acknowledges Plaintiff’s argument that valuation of a start-up can be difficult to quantify, though does not establish revenue as an appropriate input for valuation. Outside of

judicial use, the times-revenue formula is well-trodden ground for comparing valuation of pre-profit start-ups in the same field, but is not generally accepted as an appropriate way to establish objective valuation without the use of a multiplier based on industry. Daubert, however, rejected general acceptance as the sole basis to assess a methodology, replacing it with a variety of factors the court may consider. Schneider, 320 F.3d at 405 (citing Paoli, 35 F.3d at 742 n.8 (internal citations omitted)). Here, the qualifications of the expert, the non-judicial use of revenue to establish valuation, and the close relationship of Ms. Bennis' methodology to a methodology established to be reliable (the calculation of head-start damages using time value using profit or valuation as an input)¹ all make it more likely than not that Ms. Bennis' methodology is reliable under the meaning of Federal Rule of Evidence 702.

Defendants' argument that Ms. Bennis errs by including in her revenue calculation all of Akoustis' historical revenue raises a question for the jury, not a basis for exclusion under Rule 702. (See ECF No. 485.) In forming her opinion that all revenue should be included in the damages calculation, Ms. Bennis incorporates the opinion of Dr. Shanfield in finding that "the trade secrets Akoustis [are] alleged to have misappropriated related to virtually every aspect of its business and ability to get to market in the timeframe it did [including] its ability to earn grant money, fabricate products, provide engineering services, and manufacture products[,]" and that the Akoustis products were brought to market 55 months earlier than they would have without the alleged trade secrets. (ECF No. 457 at PageID 20011.) This opinion is not directly expressed in Dr. Shanfield's report, but is consistent with his report and Plaintiff's theory of the case. Ms. Bennis plainly explains her assumption: that all of Akoustis' historical

¹ Indeed, Ms. Bennis' methodology is so close to an established one that it could be characterized as a "flaw in the application" of a reliable methodology. See United States v. Gipson, 383 F.3d 689, 696-97 (8th Cir. 2004); United States v. Shea, 211 F.3d 658, 668 (1st Cir. 2000).

revenue depended on the theft of Qorvo's trade secrets. Given the scope of the trade secret misappropriation allegations at issue in this case, this assumption is not unreasonable, but if not borne out in proof, the jury will be able to dismiss Ms. Bennis' conclusions. As a result, concerns on which elements of revenue are properly included in Ms. Bennis' calculation of head-start damages are more appropriately examined in cross-examination. See Intellectual Ventures I LLC v. Check Point Software Techs., Ltd., 215 F. Supp. 3d 314, 323-324 (D. Del. 2014).

Because Plaintiffs have shown by a preponderance of the evidence that Ms. Bennis' methodology is reliable under Federal Rule of Evidence 702, and because the assumption that all revenues are proper inputs is properly challenged on cross but not at the Daubert phase, Defendants' Motion to Exclude Ms. Bennis' head-start damages opinions is **DENIED**.

ii. *Poaching*

Defendants argue that Ms. Bennis' opinions on damages for employee poaching is also unreliable because Ms. Bennis claims to calculate actual damages Qorvo suffered in replacing employees lost to Akoustis' "poaching[.]" but did not use Qorvo data on 1) whether employees were actually replaced and 2) the cost of replacement, instead relying on a report generated by a third-party assessing, in 2021, the replacement cost of employees in a company Akoustis was acquiring. (ECF No. 457 at PageID 20010.)

The trial judge has "the task of ensuring that an expert's testimony both rests on a reliable foundation and is relevant to the task at hand." Daubert, 509 U.S. at 597. "Fit" determines whether "the proffered connection between the scientific research or test result to be presented and particular disputed factual issues in the case" is sufficiently strong. Paoli, 35 F.3d at 743. Here, it is undisputed that Akoustis and Qorvo are companies of different sizes.

It is likewise undisputed that the alleged “poaching” behavior began before 2021. Despite this, Ms. Bennis uses a report which estimates the replacement cost of employees not at Qorvo in the years in question, but at a third-party company acquired by *Akoustis*. Ms. Bennis’ report does not analyze the reasons that this data would be relevant to determining the replacement costs of an employee at a much larger company over the course of the prior decade, nor does Exhibit 4, Schedule 2.1 of her report, which summarizes the allegedly “poached” employees, include date data. (See ECF No. 458.) Neither Ms. Bennis’ report nor Plaintiff’s brief provides argument as to why the use of data from a company of different size to assess the replacement cost of employees in the decade prior would be appropriate. Neither Ms. Bennis nor Plaintiffs have addressed why Qorvo’s own replacement data was not used. As such, Plaintiff cannot on the current record be said to have met their burden to show by a preponderance of the evidence that Ms. Bennis’ opinions are based on sufficient facts and data, or reliably applied to the facts of the case. Parties are therefore **ORDERED** to submit additional briefing of no more than five pages, no later than **Thursday, May 2, 2024 at 11:00 a.m.**, on these issues.

b. *Carolyn Irwin*

Plaintiff argues that Carolyn Irwin’s opinions on “avoided costs” are “not relevant . . . fail to apply accepted principles and methods for calculating [avoided costs] damages . . . [and] cannot aid the jury in calculating Qorvo’s damages.” (ECF No. 452 at PageID 18861.) Plaintiff’s central argument is that “because the avoided cost methodology requires an assumption that trade secrets were stolen and because the inputs Ms. Irwin received from Drs. Darveaux and Lebby² assume no misappropriation of trade secrets occurred,” her testimony cannot be relevant or reliable. (ECF No. 452 at PageID 18865.) Plaintiff contends that “her

² Irwin’s testimony primarily relies on the opinions of Dr. Darveaux. Those opinions which rely on excluded opinions of Dr. Lebby must also be excluded.

opinion fundamentally relies upon an assumption that Akoustis did not misappropriate any Qorvo trade secret.” (Id. at PageID 18867.) Plaintiff also argues that because Ms. Irwin admitted in her deposition that if the jury disagreed with Drs. Lebby’s and Darveaux’s opinions that certain trade secrets were capable of being assembled from public information her methodology would cease to be relevant, her methodology does not “fit” with the facts of the case.

Defendants respond that Plaintiff’s arguments take too narrow a view of trade secret law. (ECF No. 485 at PageID 23394.) They state that the reports on which Ms. Irwin based her avoided costs figures included opinions “on the time and cost for Akoustis to extract Qorvo’s purported trade secrets from publicly available sources.” (ECF No. 485 at PageID 23396.) They further argue that Ms. Irwin’s methodology is not incompatible with an assumption of liability. (Id.)

Damages experts in trade secret misappropriation cases must assume that the defendant is liable for misappropriation of a trade secret. See FieldTurf USA, Inc. v. Tencate Thiolon Middle East, LLC, No. 4:11-cv-50, 2023 WL 12290896 (N.D. Ga. Dec. 5, 2013). That Ms. Irwin did so is not disputed. Instead, Defendants argue that “Ms. Irwin’s opinions are inconsistent with the assumption that misappropriation occurred.” (ECF No. 452 at PageID 18868.) They are not. As Ms. Irwin summarized at her deposition, her calculations were not based on Drs. Lebby’s and Darveaux’s conclusions, but on the expert work they used to reach those conclusions.³ (See ECF No. 485 at PageID 23399 (quoting ECF No. 455).) This work,

³ See ECF No. 455 Ex. F. at 59:18-60:5. (“[B]oth [D]octors Darveaux and Lebby have the opinions that they’re not trade secrets, but I’m assuming they are trade secrets. I’m not operating under an assumption that these are not trade secrets. I’m setting their opinions aside and then relying on their expert work identifying the information contained in the documents that were asserted as trade secrets, finding that information contained in the documents that were asserted as trade secrets, finding that information and the estimates for implementing the methodology that was contained in that information.”)

which examines the number of hours an engineer would need to assemble or discover a trade secret from publicly available data, is not incompatible with a finding of trade secret protection—“a combination of characteristics and components, each of which, by itself, is in the public domain” can be a protectable trade secret. Life Spine, Inc. v. Aegis Spine, Inc., 8 F.4th 531, 540-41 (7th Cir. 2021). While the Third Circuit has recognized that “the DTSA . . . excludes reverse engineering from the type of conduct it defines as misappropriation[,]” their statement went to the definition of *misappropriation*, not the definition of trade secret. See Mallet & Co. Inc. v. Lacayo, 16 F.4th 364, 388, n.31 (3d Cir. 2021) (citations omitted).

Dr. Darveaux’s opinion provides hours estimates for the time an engineer would need to discover an alleged trade secret based on publicly available information. That these hours estimates are relatively low does not suffice to render them an insufficient basis for Ms. Irwin’s opinions. A compilation, even a simple one that takes little time to assemble, can qualify for trade secret protection. Even if Drs. Lebby and Darveaux found that it would take no time for an engineer to assemble certain alleged trade secrets (i.e. that they were immediately publicly available without work compiling), the hours estimates provided for others would help the jury determine the “avoided costs” for assembling those trade secrets. Unlike Edwards Lifesciences Corp. v. Meril Life Sciences PVT. Ltd., 2021 WL 5407316, at *3 (N.D. Cal. Nov. 18, 2021), Ms. Irwin does not contradict her own opinions, but rather bases her opinions on facts developed by other experts that led them to other conclusions.

Because Ms. Irwin’s use of Drs. Lebby’s and Darveaux’s data is not inconsistent with an assumption of trade secret liability, and will be helpful to the jury in understanding “avoided costs” for at least some of the alleged trade secrets, Plaintiff’s Motion is **DENIED**.

IV. CONCLUSION

For the reasons discussed above, Plaintiff's Motion to Exclude Expert Opinions of Carolyn Irwin is **DENIED**. Defendants' Motion to Exclude Expert Opinions of Melissa Bennis on "head-start" damages is **DENIED**. Parties are **ORDERED** to submit, **no later than 11:00 a.m. on May 2, 2024**, additional briefing of no more than five pages on whether Ms. Bennis' poaching opinions are based on sufficient facts and data, or reliably applied to the facts of the case given the disparity in size and timeline between the KPMG report used and the company and times in question.

IT IS SO ORDERED, this 30th day of April, 2024.

/s/ Jon P. McCalla
JON P. McCALLA
UNITED STATES DISTRICT JUDGE